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A Cognitive Two-Tier Architecture for Automated Threat Detection and Contextual Response using Agentic AI

Capstone Project Proposal

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Abstract

In the context of modern cybersecurity, Security Operations Centers (SOCs) face the critical "Big Data Paradox": an overwhelming volume of events per second leads to severe alert fatigue, with a high percentage of false positives. Traditional Intrusion Detection Systems (IDS) and Security Information and Event Management (SIEM) platforms analyze data sources in isolation, significantly reducing the ability to detect sophisticated multi-stage Advanced Persistent Threats (APT). Recently, the direct application of Large Language Models (LLMs) to log analysis has been explored; however, it introduces two critical vulnerabilities: excessive reasoning latency incompatible with real-time data streams, and the risk of LLM manipulation through Prompt Injection hidden within log fields.

To address these challenges, this study proposes a novel cognitive architecture that integrates a deterministic Rule Engine with a Large Language Model. By utilizing a two-tier approach, the system is expected to filter high-volume noise at wire speed (Tier 1) while reserving deep reasoning, powered by a LangGraph-based agent with Dual-RAG (MITRE ATT&CK and ISO 27001), for complex and ambiguous threats (Tier 2). Furthermore, a Delimited Data Encapsulation mechanism is introduced to neutralize adversarial prompt injections without sacrificing necessary contextual payload data.

The proposed system will be implemented and evaluated on benchmark network datasets (e.g., CICIDS2017). Evaluation metrics will encompass Classification (F1-Score), Operational Latency, Robustness against adversarial attacks, and Context Quality via LLM-as-a-Judge methodologies.

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1. Introduction and Background

1.1 Research Problem

Modern Security Operations Centers (SOCs) are overwhelmed by the sheer volume of security events. According to recent industry reports, the average SOC analyst processes thousands of alerts daily, with nearly half identified as false positives ([Ponemon Institute, 2023](#)). This phenomenon, known as *alert fatigue*, leads to critical threats being overlooked. Traditional IDS/SIEM systems exacerbate this problem by analyzing data sources (Web logs, Firewall logs, Authentication logs) in isolation, severely reducing the ability to detect sophisticated multi-stage attack chains such as Advanced Persistent Threats (APTs).

The application of Large Language Models (LLMs) to automated log analysis introduces two fatal vulnerabilities. First, *reasoning latency* is prohibitively high for real-time streaming environments. Second, LLMs are highly susceptible to *Prompt Injection attacks* hidden within log data ([Greshake et al., 2023](#)). Adversaries embed malicious instructions in dynamic fields (e.g., User-Agent, HTTP Payload) that the LLM processes as part of its input context. This creates a fundamental paradox: analysts must preserve these dynamic variables to analyze the attack, yet preserving them exposes the LLM to hijacking.

1.2 Research Objectives

Motivated by these limitations, this study aims to develop an autonomous AI Security Agent utilizing a cognitive two-tier architecture. Instead of relying purely on a single LLM to process all logs, the thesis proposes a hybrid approach: a high-speed deterministic rule engine combined with a deep-reasoning LLM Agent. This approach is expected to drastically reduce reasoning latency by filtering noise, mitigate prompt injection via bounded encapsulation, and enhance context awareness using Long-Term Threat Memory and a Dual-RAG setup (incorporating MITRE ATT&CK and ISO 27001). A Human-in-the-Loop (HITL) Web Dashboard will also be developed to demonstrate practical SOC automation.

1.3 Research Questions

Based on the objectives, this thesis focuses on four core research questions: 1. Does a Two-Tier Architecture (Rule-based + LLM Agent) combined with Session-Aware Behavioral Baselineing optimize Reasoning Latency compared to a One-Tier (LLM-only) approach? 2. Can a Delimited Data Encapsulation mechanism (using Dynamic Randomized Delimiters) effectively prevent structural Prompt Injections (e.g., Delimiter Smuggling) without degrad-

ing the LLM's classification accuracy? 3. How does a Dual-RAG pipeline (combining tactical and defensive frameworks) improve the Context Relevance of the LLM's threat analysis? 4. Can an adaptive Feedback Loop with HITL Quarantine prevent Adversarial Rule Injection and safely automate defensive responses?

1.4 Scope of the Study

The research is limited to Network Intrusion Detection and SOC log triage. The in-scope components include: Data Streaming Pipeline (Redis), Tier-1 Rule Engine, LangGraph Agent with Dual-RAG, Guardrails (Volume Compression and Injection Defense), and a Streamlit HITL Dashboard with SQLite-based Threat Memory. The study does not cover fine-tuning Foundation Models, physical hardware deployment (e.g., physical Kafka clusters), or Layer 3/4 volumetric DDoS mitigation.

1.5 Thesis Structure

The thesis will be organized into six main chapters: (1) Introduction and Background; (2) Literature Review; (3) Proposed Solution and Research Methodology; (4) Implementation and Evaluation Plan; (5) Expected Results and Contributions; and (6) Conclusion.

2. Literature Review

2.1 Foundational Approaches

In the early stages of network security, Intrusion Detection Systems (IDS) primarily relied on signature-based matching (e.g., Snort, Suricata) to detect known threats. While highly efficient in terms of throughput, these systems exhibit significant limitations in identifying zero-day attacks and polymorphic malware ([Khraisat et al., 2019](#)).

The advent of Machine Learning (ML) led to anomaly-based IDS, where models learn a baseline of normal network behavior and flag deviations. While Random Forests and Support Vector Machines (SVM) improved zero-day detection, they generated high false-positive rates due to the dynamic nature of benign network traffic. To address these limitations, modern Security Information and Event Management (SIEM) systems aggregate logs, yet they still require extensive manual triage by human analysts to construct the full context of an attack.

2.2 Related Work

The recent emergence of Large Language Models (LLMs) has opened new directions for automated log summarization and incident response. Studies such as those by [Oniagbi et al. \(2024\)](#) demonstrate the use of cloud-based APIs (e.g., GPT-4) for SOC automation. However, relying on cloud APIs violates strict organizational data privacy policies and introduces unacceptable network latency for real-time triage.

In the realm of Retrieval-Augmented Generation (RAG) for cybersecurity, frameworks like RAG-ATT&CK ([RAG-ATT&CK, 2024](#)) have applied knowledge graphs for threat intelligence. Yet, existing studies generally focus solely on tactical analysis, lacking the integration of defensive compliance standards (such as ISO 27001) required for actionable SOC remediation.

Regarding adversarial robustness, the Agent Security Bench (ASB) ([Zhang et al., 2024](#)) highlights that LLM agents suffer from high attack success rates via prompt injection. While the Drain3 algorithm ([He et al., 2017](#)) is widely used for log template mining, it is typically employed for storage compression rather than active injection defense. These studies provide essential scientific support for the necessity of a unified, self-defending architecture, revealing a critical research gap that this thesis seeks to address.

3. Proposed Solution and Research Methodology

3.1 Proposed Solution: Cognitive Two-Tier Architecture

The proposed solution is based on a cognitive Two-Tier Architecture designed with a strict Separation of Concerns. The core idea is to leverage the speed of deterministic rules (Tier 1) alongside the deep reasoning capabilities of an Agentic LLM (Tier 2), interconnected by robust adversarial guardrails.

At the first stage, the Tier 1 Rule Engine acts as a heuristic filter. It maintains Session-Aware Behavioral Baselines using in-memory data structures (Redis) to track request rates and known malicious IPs. This layer is designed to drop over 90% of benign or purely volumetric noise at wire speed, ensuring the downstream LLM is not overwhelmed.

Logs that exhibit anomalous behavior or trigger ambiguous thresholds are passed to the Guardrail Layer. Here, a two-step neutralization occurs: 1. **Template Mining (Drain3)**: Extracts the static log template to compress the token volume without altering the dynamic payload variables. 2. **Delimited Data Encapsulation**: Dynamic payload fields are wrapped within cryptographically generated randomized delimiters (e.g., <<RANDOM_HEX>>). This prevents adversaries from using Delimiter Smuggling to escape the data context and execute Prompt Injections.

Following the guardrails, the sanitized log enters Tier 2, governed by a LangGraph-based Agent. The Agent utilizes a Dual-RAG mechanism, querying a local FAISS vector database embedded with MITRE ATT&CK tactics and ISO 27001 controls. The LLM synthesizes this knowledge to determine whether the event is malicious.

Finally, to ensure system safety against Adversarial Rule Injection, the LLM cannot directly modify the Tier 1 firewall. Instead, its recommended rules are placed into a Quarantine state (Human-in-the-Loop Feedback Loop). A SOC manager reviews these rules via a Dashboard before they are hot-reloaded into the Tier 1 engine.

3.2 Technologies and Tools

The system is built on modern Python frameworks. The data streaming and short-term session tracking rely on Redis. The LLM reasoning engine utilizes local quantized models (e.g., Gemma-2-9B-IT) served via the Oobabooga text-generation-webui API, ensuring absolute data privacy.

The agentic workflow is orchestrated by LangGraph, enabling cyclical decision-making and tool-use. For knowledge retrieval, `sentence-transformers` is used alongside FAISS

for efficient semantic vector search. The user interface and Human-in-the-Loop dashboard are developed using Streamlit, while long-term threat memory and audit trails are persisted using SQLite.

4. Implementation and Evaluation Plan

4.1 Main Development Stages

The implementation process will first focus on the Data Ingestion and Tier 1 Rule Engine. Standard network security datasets, such as CICIDS2017 or UNSW-NB15, will be preprocessed and streamed to simulate real-world log generation. The session-tracking mechanism will be built using Redis TTLs to establish behavioral baselines.

Following this, the Guardrail Layer will be implemented, integrating the Drain3 algorithm for template mining and developing the cryptographic Delimited Data Encapsulation module to neutralize adversarial payloads.

The next critical step is the construction of the Tier 2 Agent using LangGraph and the Dual-RAG knowledge base. The FAISS index will be populated with parsed documents from MITRE and ISO standards. The agent's prompts will be carefully engineered to strictly separate system instructions from the encapsulated payload data.

Finally, the Streamlit Dashboard and the SQLite Long-Term Threat Memory will be integrated. This phase will finalize the closed-loop process by enabling the Human-in-the-Loop feature, allowing administrators to approve dynamic rules generated by the LLM and feed them back to Tier 1.

4.2 Evaluation Plan

To validate the effectiveness of the system, the study defines an extensive evaluation strategy termed the 4D (or 5D) Evaluation Framework.

To evaluate the system's baseline accuracy, an Ablation Study will be conducted. This will compare the full Two-Tier system against a baseline LLM-only setup to observe differences in performance. The evaluation metrics will include:

- **Classification Metrics:** F1-Score, Precision, Recall, and False Positive Rate (FPR), analyzed using McNemar's Test for statistical significance.
- **Operational Metrics:** Reasoning Latency (proxy for Mean Time To Detect) and Semantic Cache Hit Rate, evaluated using the Mann-Whitney U Test.
- **Robustness Metrics:** The Guardrail Defeat Rate will be tested against curated adversarial datasets containing Structural Attacks (Delimiter Smuggling) and Semantic Confusion.

- **Context Quality Metrics:** To evaluate the LLM's reasoning, an LLM-as-a-Judge methodology (inspired by the RAGAS framework) will assess Context Precision, Answer Relevancy, and Faithfulness.

5. Expected Results and Contributions

5.1 Expected Results

This study is expected to develop a complete, autonomous pipeline capable of performing the full sequence of SOC tasks—log ingestion, heuristic filtering, adversarial sanitization, deep reasoning, and rule generation.

From a technical perspective, the Two-Tier architecture is anticipated to significantly reduce the overall processing latency compared to standalone LLM inference, as the Tier 1 engine will filter out the vast majority of benign traffic. The integration of Delimited Data Encapsulation is expected to achieve a high blocking rate against structural prompt injection attacks, demonstrating a robust defense mechanism without stripping away the necessary payload forensics required by the LLM.

Furthermore, the Dual-RAG integration is expected to yield high scores in Answer Relevance and Faithfulness, ensuring that the Agent generates actionable, compliance-aligned remediation steps rather than hallucinated responses.

5.2 Research Contributions

From an academic standpoint, the primary contribution of this research is the proposal of a novel Cognitive Two-Tier Architecture that securely bridges deterministic rule engines with Agentic LLMs. This approach remains underexplored in cybersecurity literature, particularly concerning the deployment of local, privacy-preserving LLMs equipped with dynamic adversarial guardrails.

The second contribution lies in the methodological framework for evaluating LLM-based SOC agents. By introducing the 4D/5D Evaluation Framework—combining statistical classification tests, operational latency metrics, and LLM-as-a-Judge context quality assessments—this study provides a comprehensive baseline for future research in AI-driven cybersecurity.

Finally, the practical contribution is the establishment of a standardized, open-source pipeline (SENTINEL) that integrates all stages of alert triage and Human-in-the-Loop remediation. This serves as a scalable, real-world framework that organizations can adapt to alleviate SOC alert fatigue safely.

6. Conclusion

In this study, we propose a Cognitive Two-Tier Architecture that combines a high-speed deterministic Rule Engine with a deep-reasoning Agentic Large Language Model. This architecture is designed to address the "Big Data Paradox" and alert fatigue currently plaguing modern Security Operations Centers, while simultaneously mitigating the inherent vulnerabilities of applying LLMs directly to adversarial log data.

The system leverages the speed of Tier 1 to drop massive volumes of noise and relies on robust Guardrails—specifically Delimited Data Encapsulation—to neutralize prompt injections before they reach the Tier 2 Agent. Supported by a Dual-RAG knowledge base consisting of MITRE ATT&CK and ISO 27001 standards, the Agent can provide highly contextualized, accurate threat assessments. Finally, the closed-loop design ensures safety through a Human-in-the-Loop Quarantine system.

The expected contributions of this research include proposing a novel hybrid architecture, providing empirical evaluation methodologies for AI Security Agents, and developing a practical, scalable pipeline for real-time automated incident response. Future research directions may investigate multi-agent orchestration for distributed network topologies or the integration of multi-modal log analysis.

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